Calibration complete; validation is separate

Evidence checked 2026-10-05 21:57:51 CEST (19:57:51 UTC)

352 / 352 calibration archives verified

345 paired runs completed normally; 7 numerical-failure pairs were archived and retained. After verifying repeated endpoints and deduplicating: 240 Quiroga outcomes (233 complete, 7 failed) and 120 FRAMES outcomes (all complete). No candidate passed the full calibration gate.

Independent settings now frozen

Quiroga: fixed alpha_Q = 0.5, lambda_Q = 128 × alpha_Q = 64; adaptive-step multiplier = 0.01. FRAMES: alpha_F,0 = 5e-05, beta_0 = 10000, p = 0.25; alpha_F,k = alpha_F,0 × (k + 1)^0.25 increases during optimization.

Selected-candidate calibration endpoints

Metric	Quiroga	FRAMES	Gate threshold
Process fidelity	0.062815927 0 / 8 pass	0.999961452 22 / 24 pass	≥ 0.99
Full-design RMSE	0.051629 0 / 8 pass	0.00036692 0 / 24 pass	≤ 1e-6
Normalized TP error	0.00043225 8 / 8 pass	0.0046325 4 / 24 pass	≤ 0.001
Full recovery gate	0 / 8 pass	0 / 24 pass	All three

Medians are over 8 channel means at 2,000 row epochs. FRAMES averages its 3 sampling streams within each channel first. Pass counts are native outcomes: 8 deterministic Quiroga runs versus 24 FRAMES streams—not 24 independent channels. Both use 256,000 nominal row accesses per run; this is not an equal-compute comparison.

What the calibration does—and does not—show

The predeclared order was full-gate rate, then number of gate components passed, then fidelity, RMSE and TP tie-breakers. With no full-gate passes, Quiroga's TP-feasible settings ranked ahead of higher-fidelity, non-TP settings. The selected Quiroga setting is not its maximum-fidelity setting: that candidate reached median fidelity 0.105339 but passed no gate components.

FRAMES has high normalized process fidelity here, but its median TP error and RMSE still miss the gate. Q alpha is at the upper grid boundary; F alpha0 is at the lower boundary. These are best settings within the declared grid—not global optima, successful recovery, or an overall winner. Findings are specific to this pinned implementation, sensing, initialization, grid and budget; they do not establish that Quiroga's published method fails. Benchmark readiness is not yet established.

Validation plan and current run snapshot

Separate channels; frozen choices; no reranking

Confirmatory design frozen before validation

24 paired runs: 8 new channels (100–107), one shared initialization per channel and 3 FRAMES sampling streams. Data, initial factor, rank and physical gate are shared; TP penalties are not.
Budget: 10,000 row epochs = 1,280,000 nominal row accesses per method; 10,000 Quiroga updates and 40,000 FRAMES updates. Checkpoints every 100 row epochs.
No selection or retuning on validation outcomes. Held-out channels remain untouched.

Scheduler / artifact snapshot

Ruche array job: 2014377 | State: RUNNING: channel 102; five later channels pending at JobArrayTaskLimit
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Archived paired trials: 8 / 24 | Archived numerical-failure pairs: 5
Per task: cpu_short, 2 CPUs, 1 GB, 20 minutes; array concurrency 1.

Channels 100 and 101 have finished all three paired archives each; channel 102 has archived two trials and is still running. Three pairs completed normally and five retain numerical method failures. Slurm FAILED/exit2 denotes retained method failure, not missing data. Repeated deterministic Quiroga outcomes must be deduplicated within each channel. This is a live partial count, not the final validation result.

First downloaded validation trial—not an aggregate

qpt-val-n2f-c100-i0-s0-20261005-2136

Method	Status	Fidelity	RMSE	TP error	Gate
Quiroga	failed	unavailable	unavailable	unavailable	fail
FRAMES	complete	0.999999989	1.8061e-05	0.00019848	fail

Quiroga failed numerically after 8 updates; the failed endpoint and valid FRAMES result were preserved. FRAMES passes fidelity and TP, but misses the RMSE threshold. A single trial cannot establish the validation pass rate. A Slurm exit 2 can mean archived numerical failure, not missing artifacts; failed outcomes remain counted.

Reproducibility / preserved evidence

Source: c7a8c9789b78
/home/tony/frankwolfe

Selection: fa9b5d5287f26029
/home/tony/qpt_quiroga_continuation_20261005_2136/selection_n2_full/selection_manifest.json

Frozen validation plan: 2704908ff5c019ab
/home/tony/qpt_quiroga_validation_20261005_2136/validation_plan.json

Independent audit: cc6a394d0faf9c08
/home/tony/qpt_quiroga_continuation_20261005_2136/independent_calibration_audit.json

Earlier aborted runs remain preserved. The c7a8c97 archival repair did not change optimizer equations, penalties or gate thresholds. Checks: 9 selector regression tests and 13 validation wrapper tests passed, with independent reviews.